

Review

A Comprehensive Review of Artificial Intelligence (AI) Companies in the Power Sector

Vladimir Franki ^{1,2,*} , Darin Majnarić ³ and Alfredo Višković ^{1,2}¹ Faculty of Engineering Rijeka, University of Rijeka, Vukovarska 58, 51000 Rijeka, Croatia² Energy Platform Living Lab, Unska 3, 10000 Zagreb, Croatia³ Faculty of Mechanical Engineering and Naval Architecture, University of Zagreb, Ul. Ivana Lučića 5, 10000 Zagreb, Croatia

* Correspondence: vfranki@riteh.hr; Tel.: +385-51-651444; Fax: +385-51-651416

Abstract: There is an ongoing, revolutionary transformation occurring across the globe. This transformation is altering established processes, disrupting traditional business models and changing how people live their lives. The power sector is no exception and is going through a radical transformation of its own. Renewable energy, distributed energy sources, electric vehicles, advanced metering and communication infrastructure, management algorithms, energy efficiency programs and new digital solutions drive change in the power sector. These changes are fundamentally altering energy supply chains, shifting geopolitical powers and revising energy landscapes. Underlying infrastructural components are expected to generate enormous amounts of data to support these applications. Facilitating a flow of information coming from the system's components is a prerequisite for applying Artificial Intelligence (AI) solutions in the power sector. New components, data flows and AI techniques will play a key role in demand forecasting, system optimisation, fault detection, predictive maintenance and a whole string of other areas. In this context, digitalisation is becoming one of the most important factors in the power sector's transformation process. Digital solutions possess significant potential in resolving multiple issues across the power supply chain. Considering the growing importance of AI, this paper explores the current status of the technology's adoption rate in the power sector. The review is conducted by analysing academic literature but also by analysing several hundred companies around the world that are developing and implementing AI solutions on the grid's edge.



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1. Introduction

Digitalisation is fundamentally transforming the world we live in. It has a growing impact on society, affecting people both at national and international levels [1]. Nationally, digitalisation and the accompanying process of automation are likely to cause a shift from low-skilled to high-skilled occupations [2]. Internationally, similar to the shift towards renewable energy, it may alter the delicate power balance among nations. Digitalisation has been affecting both our personal and professional lives for years and will continue to do so with even stronger implications in the future [3]. Similarly, the energy sector has been going through a transformation of its own [4]. After the dissolution of vertically integrated utility business models [5], the energy sector has been presented with a new set of challenges regarding environmental issues and the surge of renewable energy sources [6]. The phase of reaching a zero-carbon power system is still far from completion, yet we are now witnessing a dawn of a new era [7]. It is hard to define how this new period might be labelled, but what's certain is that it will be shaped by the progress of digitally-enabled solutions [8]. During the following years, digitalisation will be a key facilitator of the new energy transformation [9]. The energy system is a complex behemoth consisting of countless elements, all working together to provide a reliable supply of energy in

real-time. In the future, this system will become even more complex as the traditional centralised flow of energy is disrupted by the emergence of new distributed sources [10]. Electric vehicles (EVs), battery systems, smart appliances and the electrification of buildings will put additional strain on the distribution part of the power supply chain by adding new components, introducing new market players and facilitating bi-directional flows of energy [11].

The intricacies that arise when trying to maintain the delicate supply-demand balance will require the utilisation of complex, automated algorithms that are able to optimise the system in real-time [12]. In order to make informed decisions, these algorithms will need to process vast amounts of data. In this sense, the relatively recent rise of supercomputing and Big Data has empowered the development of various solutions based on Artificial Intelligence (AI) [13]. As the power supply chain gradually becomes more complex, the application of automated, advanced techniques in system management will become a necessity [14]. It is important to note that this will not be an unimaginable leap as techniques such as artificial neural networks (ANNs), reinforcement learning (RL), genetic algorithms (GAs) and multi-agent systems are all commonly explored as tools in forecasting, optimisation and control of the power system [15]. Currently, due to the lack of advanced automation infrastructure, a number of system operations are still performed at basic levels of automation [16]. According to a growing number of research papers, technical reports and case studies, AI will play a crucial role in the power system of the future, limiting manual interventions and introducing advanced techniques of system optimisation [17]. At present, most papers study the application of AI in the process of power flow optimisation. In this sense, AI will essentially be used to convert data gathered from the power system into useful information, which will then be used to formulate concrete steps in optimising system operation. However, as evidenced by various research papers and numerous start-ups, AI's role in the power system will not be limited to power flow optimisation. Areas such as fault detection, predictive maintenance, demand response, forecasting and customer relations all have significant potential for AI implementation.

Motivation, Related Works and Scope

AI-based solutions are increasingly utilised across various segments of the power sector. The expanding interest in AI can also be evidenced by the surge of research interest in the field. The aforementioned trend necessitates the need to create a systematic overview of the application fields of AI in the power sector. In this context, there are a number of review papers that explore AI approaches to solving issues related to the power system. Ahmad et al. provide a comprehensive review of AI in energetics systems identifying seven key areas of its application [18]. In another paper, Ahmad et al. explore recent advances in AI technology and focus on a narrower application of AI in balancing supply and demand and generating power from solar and hydrogen [19]. Considering papers with slightly higher impact factors, we identified another two comprehensive reviews that look into the entire power sector. Ali et al. analysed AI techniques supporting distributed smart grids [16], while Kumbhar et al. focused on machine learning (ML) applications in the smart grid and the renewable energy sector [20]. Given the large number of areas in which AI can aid the power sector, the majority of reviews focus on a certain aspect of potential AI application rather than taking a holistic approach to the matter. Antonopoulos et al. focus on ML techniques used in demand response [21], Zhang et al. provide a review on the deep learning (DL) approach in frequency analysis and system operation management [22], while Hou et al. [23] and Yazici et al. [24] explore demand forecasting techniques. Other notable reviews look into AI applications regarding the performance of clean energy communities [25], integration of renewable energy sources (RES) in the existing power system [26–28], supply chain automation [29], smart buildings [30], implementation of smart meter infrastructure [31], regulatory issues that AI technologies face [32] and the data AI uses [33]. Security issues of AI in the power system are a standalone field that requires a focus of its own [34,35]. Figure 1 shows the surge of research papers focused on the topic of

artificial intelligence. The figure has two parts (left and right). The left part is obtained by listing all Scopus-indexed research papers containing the keyword “artificial intelligence”. The right part is obtained by listing all research papers that contain the keywords “artificial intelligence” and “power system”.

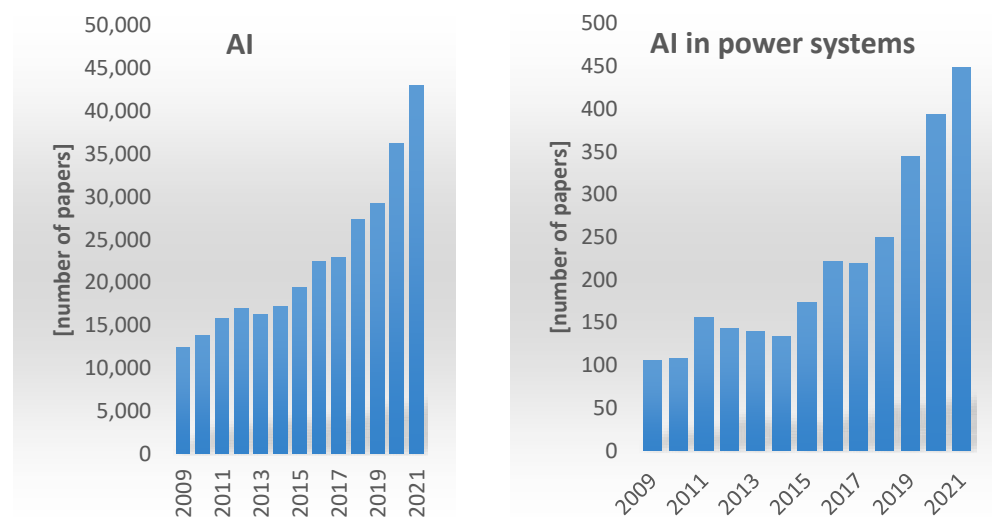


Figure 1. Research papers on AI (left) and AI in the power system (right).

Observing previously published research in the field, a significant number of papers offering reviews on AI-based techniques in the power sector can be found. Given a large number of possible areas of application, the majority of papers focus on a particular segment of AI implementation, such as predictive maintenance, demand response, forecasting energy demand, etc. Despite numerous reviews already published, there is a surprising lack of papers that aim to construct a systematic, high-level analysis of all the main fields of AI applications in the power sector. The main aim of the research presented is to establish what are the key areas of research focus regarding the implementation of AI-based solutions in the power sector. Taking it a step further, the paper aims to integrate the analysis of existing academic knowledge with real-life business applications. Naturally, there is a significant number of large, energy-based companies, such as Enel, Engie, ABB, Schneider, General Electric, Honeywell, etc., that have, in recent years, developed strong AI competencies. However, the research presented in the paper focuses only on companies whose core business is artificial intelligence. Within this framework, over 450 companies across the world were observed during the market analysis part of the research. From these companies, a list of promising start-ups and well-established companies is compiled, giving the answer to which fields of application are getting the most traction. To sum up, and against the aforementioned background, the major contribution of the paper is a systematic review of the main AI-based solutions in the power sector that integrates academic knowledge with practical applications. To the best of our knowledge, this is the most comprehensive review of AI-based companies in the power sector to this date (as far as academic papers are concerned). The following paragraphs have a two-fold goal:

Provide a comprehensive, systematic literature review of the current research in the field of AI solutions in the power sector and identify key areas of the potential application of AI technology in the field.

Taking the scientific literature review further, the paper provides a structured insight into the adoption rate of AI technologies in the power sector by observing the current efforts of established companies and start-ups.

With regards to the power sector, in this paper, we provide (1) a systematic review of the main fields of research regarding AI-based solutions and (2) a comprehensive analysis of current AI applications in real life. The introduction segment opens the topic of AI in the

power sector and reveals key aspects of the paper. Paragraph 2 reveals the method used during the analysis. Paragraph 3 offers a literature review and a systematic analysis of the main field of research regarding AI solutions in the power sector. Paragraph 4 analyses the current AI adoption rate in the power sector by observing AI companies. Finally, Paragraph 5 gives a final conclusion to the topic.

2. Method

The first part of the paper presents an academic literature review. Its aim is to outline the most researched topics regarding the application of AI-based solutions in the power sector. The process of identifying relevant literature follows a four-stage progression:

- Identification by keywords and titles. The key tool used for identifying relevant literature has been the Scopus database. There were three main criteria during the paper selection process of stage one: journal impact factor, citations and year of publication.
- Inclusion of related work. Additional documents were included based on key references in papers from stage one.
- Selection by abstract. The abstracts of each paper selected during phase one were examined.
- Selection by full text. Each selected paper has been examined. Key methods and focus areas were identified.

During the process, a number of queries were utilised. Search queries were formed by combining several of the key search terms from Figure 2. For example: “Artificial Intelligence” (or “Machine Learning” or “Deep Learning”) AND “Forecasting” AND “Demand”. The results obtained have been carefully reviewed and filtered. Our research confirmed the existence of numerous research papers analysing potential AI solutions in the power sector. As the paper focuses on the entire power sector, only a portion of them are referenced. However, the results obtained closely relate to actual research efforts and researchers’ focus areas.

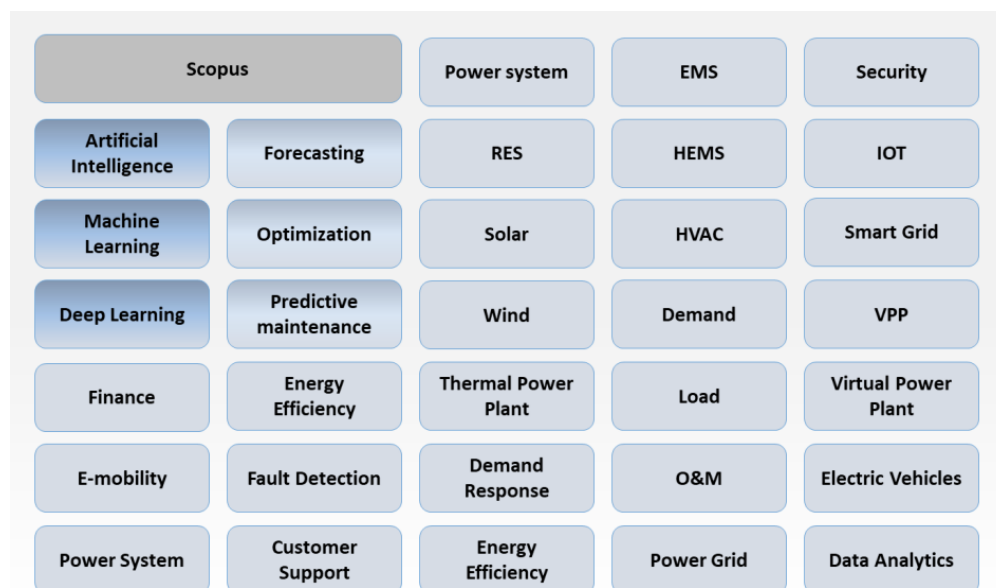


Figure 2. Search methodology and key search terms.

The second part of the paper integrates academic literature findings with a market analysis of real-life applications by firms operating in the sector. The list of companies developing AI solutions was obtained via multiple sources. Some companies are listed in other research papers, such as [21] and [36]. Some companies are listed in various databases, such as ai-startups [37], tracxn [38], Omdena [39], startus-insights [40], and crunchbase [41].

A number of companies were identified via news segments and web pages. Overall, around 500 start-ups and companies were examined during the market analysis. After disregarding inactive companies and identifying the ones whose core business is AI, a list of 220 companies was formed. To this date, and to the best of our knowledge, the list presented in Appendix A is the most extensively published in any research paper.

3. Research on AI Approaches in the Power Sector

As said by one of the bibles of AI [42], “we call ourselves Homo sapiens—man the wise—because our intelligence is so important to us”. The study of AI does not simply aim to develop an understanding of how our cognitive reasoning works but to replicate its processes in order to automate solving complex issues we face on a daily basis. Despite being considered a relatively new field in science, AI has been present in our minds for quite some time before it was established as an academic discipline in the 1950s [43] when Alan Turing published a ground-breaking article in which he described an intelligent machine and proposed a method on how to test its intelligence [44]. However, it remained a somewhat obscure area with limited practical interest up until recent times when advances in the processing power of our computers and the rise of data collection started bringing AI closer to our lives. This is an expansive, ongoing process involving a multidisciplinary approach. The fundamental ideas of AI lie in philosophy and the study of how we perceive our environment, process information and make the “right decisions”, but in order to make a step forward towards standardisation of these concepts, the introduction of logic, probability and computation is needed.

AI is a multidisciplinary domain that utilises various knowledge bases, which range from information technology and engineering to neuroscience or finance. During decades of research, multiple approaches have been used in the AI process. These approaches include but are not limited to, symbolic reasoning [45,46], statistical learning [47,48], Bayesian optimisation [49–51], soft computing [52–54], logic-based [55,56] and knowledge-based systems [57,58]. Most common, modern AI approaches in tackling power system issues can be divided into four categories [19]: machine learning (ML) [20], multi-agent systems [59,60], nature-inspired intelligence [61] and artificial neural networks (ANN) [62–64]. The term artificial intelligence is often interchanged with two other popular concepts: machine learning and deep learning (DL). Although closely related, these techniques are not indistinguishable [65]. Machine learning is a subset of AI and is characterised by a learning process without explicit programming. In other words, ML describes AI methods that can automatically identify data patterns and then use these patterns to forecast future data in an uncertain environment [66,67]. ML can be classified into four categories: supervised, unsupervised, reinforced and evolved learning [68]. Deep learning, another familiar concept in the AI family, is a branch of machine learning methods [69]. DL typically involves the implementation of large neural networks (although this is not a prerequisite) [70]. DL is used to learn multiple levels of representation and abstraction and is developed to have the ability to process raw data [71]. DL gained recognition during the 2012 ImageNet Challenge, after which DNN/CNN/RNN algorithms started to get increasingly used by developers [72]. Figure 3 reveals which of the four main AI techniques are utilised in the papers reviewed in this paper.

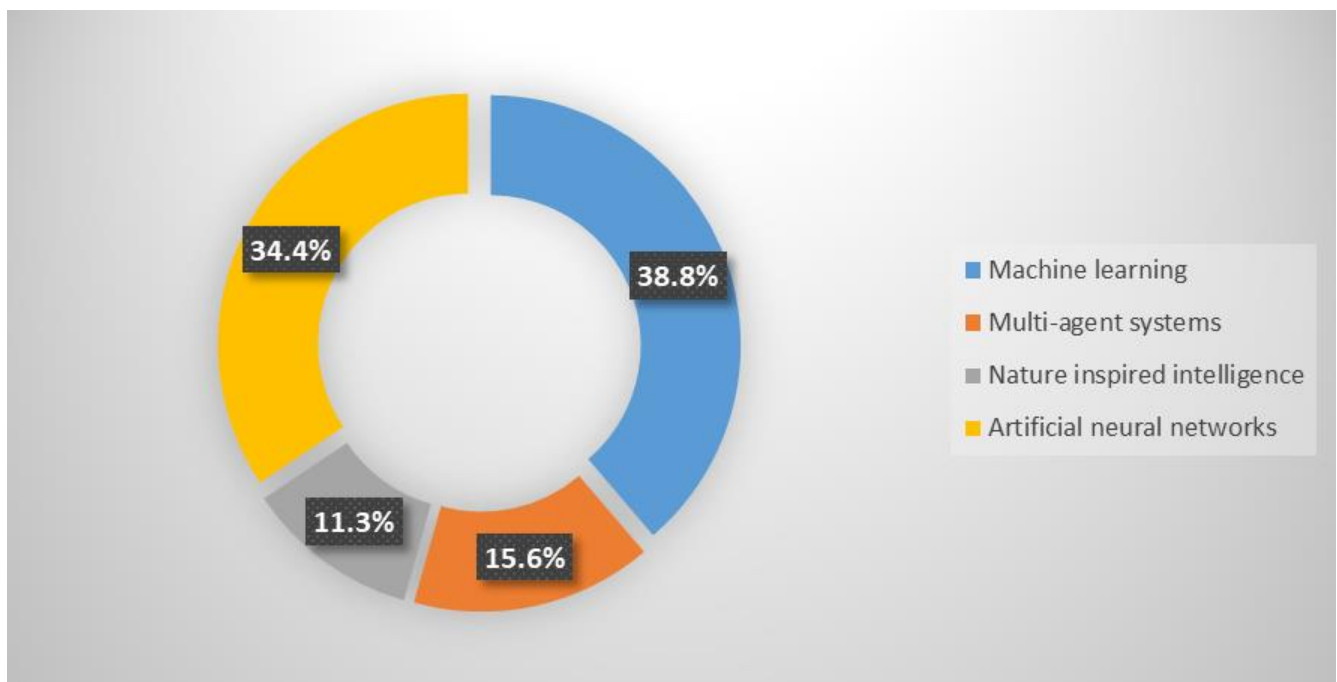


Figure 3. AI techniques used in the reviewed literature.

As briefly mentioned in the introduction segment, the power sector is currently going through a radical transformation. This transformation is mainly driven by “3Ds”: decarbonisation, decentralisation and digitalisation. In essence, all three processes are heavily linked and interdependent. The power system of the future will be based on renewables and battery storage [73]. Renewable energy relies on distributed generation. In a power system composed of a large number of distributed sources, intelligent (smart) infrastructure plays a key role in optimising resource management. Digitally-enabled solutions are, thus, critical in sustaining our decarbonisation efforts. AI is seen as an emerging technological field, an innovation front and an enabling technology [74]. Advances in computational power and AI techniques enabled AI to play a notable role in global business arenas [75]. AI technology rapidly progresses and is accompanied by a growing research presence—the power sector is not an exception. In this context, AI-aided solutions can find their place in all four segments of the energy supply chain: generation, transmission, distribution and supply [76].

3.1. Forecasting

Reliability is one of the key aspects of a successful energy supply chain. To ensure grid stability, power consumption and generation need to be in constant equilibrium. Balancing countless components with time-dependent outputs and/or demands is an extremely difficult task to manage. This is particularly emphasised when techno-economic constraints of the generation portfolio are taken into consideration. What further complicates things is the surge of variable RES. These generation capacities have intermittent outputs that are oftentimes difficult to predict and challenging to incorporate in hourly dispatch curves. In addition, the traditional energy system is based on a centralised flow of energy from large producers (typically thermal power plants) to consumers. However, the emergence of distributed energy sources and prosumers is disrupting this paradigm and introducing bidirectional power flows. In this context, the transition from a traditional power system to a decentralised, smart system of the future will necessarily entail the implementation of advanced management software, as shown in Figure 4.

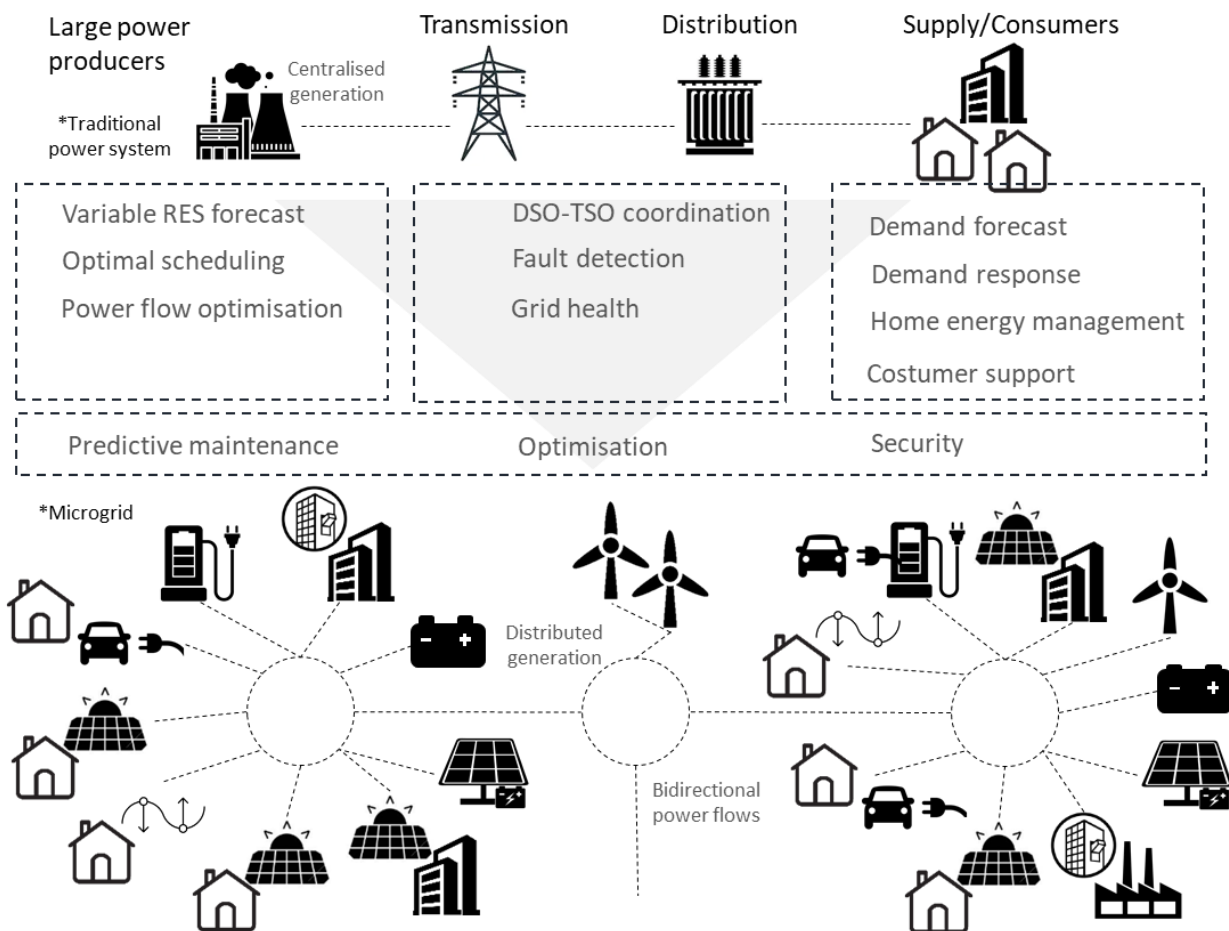


Figure 4. The AI-aided new energy paradigm.

Uncertainty regarding the variability of both demand and generation significantly increases risks and costs associated with the reliable provision of energy. In the past, the main solution to the issue of variability was to install backup generation capacities. However, the rise of digitally-enabled solutions introduces algorithms that aim to forecast the potential supply-demand curve and optimise assets to achieve optimal dispatch. In this context, forecasting has been one of the first areas of AI application in the power sector. Short-term forecasting of electricity demand is already a well-established topic and has been researched for over two decades [77,78]. Studies in short-term demand forecasting have now started to shift towards exploring smart grids and smart buildings [79]. AI techniques are also being applied in forecasting medium and long-term demand levels using different approaches such as fuzzy logic [80], multivariate adaptive regression splines (MARS), artificial neural network (ANN) and linear regression (LR) methods [81].

In addition, the process of predicting generation outputs from variable renewables is receiving an increasing amount of attention. Successful prediction of wind or solar power plant output requires the installation of smart sensors and the integration of this data with GPS-based weather forecasting platforms. AI methods have been researched regarding solar units using the adaptive network-based fuzzy inference system (ANFIS) [82] and utilising ML [83] and ANN approaches [84]. ML [85] and ANN [86] approaches are also used in forecasting wind power production. Long-term forecasts on wind power are also a topic of strong researchers' interest [87]. There are several studies that confirm that the application of AI-based algorithms is able to notably improve the accuracy of predicting wind [88,89] and solar generation [90,91]. In addition to forecasting demand or production outputs, AI techniques are also commonly explored in order to predict the prices of energy commodities [92] and electricity prices [93–95].

3.2. Optimisation

As evidenced by our research, optimisation applications in the power sector can be divided into asset optimisation and system-level optimisation. In this context, asset optimisation refers to the process of operational optimisation of a single unit, such as a power plant, a grid component, a storage system, a home energy management system (HEMS), and/or a building heating, ventilation and air conditioning (HVAC) system. There are numerous papers analysing how to augment the efficiency of variable renewables such as solar [96,97] and wind power plants [98,99]. Other papers focus on traditional power plants [100], grid assets [101] and energy storage systems [102]. Looking at the consumer side of the equation, papers focus on HVAC systems [103], energy efficiency in buildings [104] and HEMSs [105]. In addition to operational optimisation, AI-aided solutions also focus on predictive maintenance and fault detection.

Another large AI research area refers to energy resource management. System-level optimisation focuses on a broader area optimising power flows and dispatch schedules. In this context, there is a considerable amount of papers that focus on demand response (DR) [19]. Some papers focus on the demand response of households [106], some explore buildings [107], while others analyse industrial facilities [108]. In addition, there are three other, relatively novel areas where AI optimisation is researched: virtual power plants (VPPs) [109], battery systems [110] and e-mobility [111]. With the emergence of prosumers and distributed energy sources (DERs), the virtual power plant concept is quickly gaining momentum [112]. AI regarding VPPs is explored in the context of schedule optimising [113] and forecasting flexibility [114]. Considering the surge of distributed sources and variable renewables, AI is increasingly becoming an important segment of system operation and control. Figure 5 reveals a typical fault occurrence in a power grid and distinguishes how primary, secondary and tertiary control help in bringing the system back in balance. It also outlines the main field of operation for an AI-aided system that can automate and optimise the recovery process. AI approaches show a distinct advantage in solving such complex problems in real-time [115].

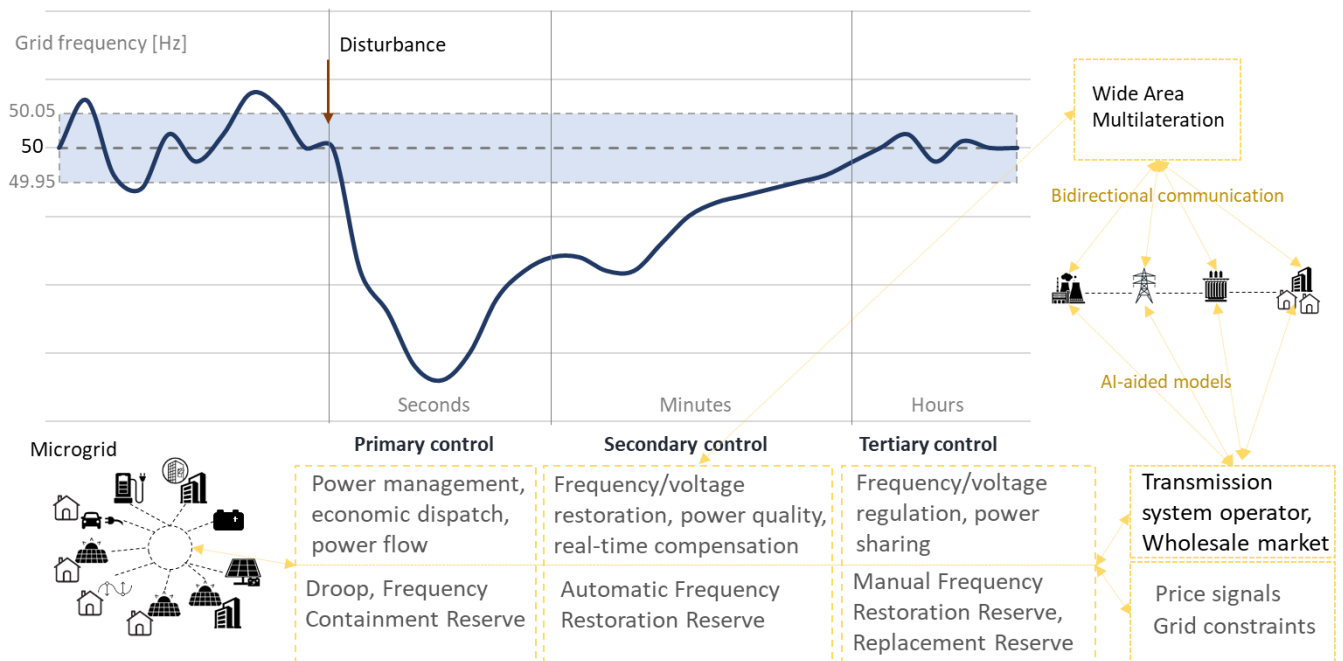


Figure 5. AI-aided fault recovery.

Regarding battery storage, AI is used to explore digital twins in management systems [116], predict novel materials with designed properties [117] and facilitate the process of searching for novel electrochemical energy storage materials [118]. E-mobility is an

important part of almost any sustainable development scenario. However, at present, it seems that the topic of e-mobility is not at the focus of the AI-based research community. In other words, there are not many papers that analyse how AI solutions can be applied to e-mobility. Research in this particular area is mostly focused on analysing how AI can help with electric vehicle (EV) mass adoption [119], how it can help with smart charging [120] and EV energy management and charging scheduling systems [121]. Taking it a step further from Figures 5 and 6 reveals the key building blocks of an AI-aided energy management system that would help optimise the operation of the energy supply chain, taking into consideration the techno-economic aspects of power generation capacities, grid constraints and forecasts.

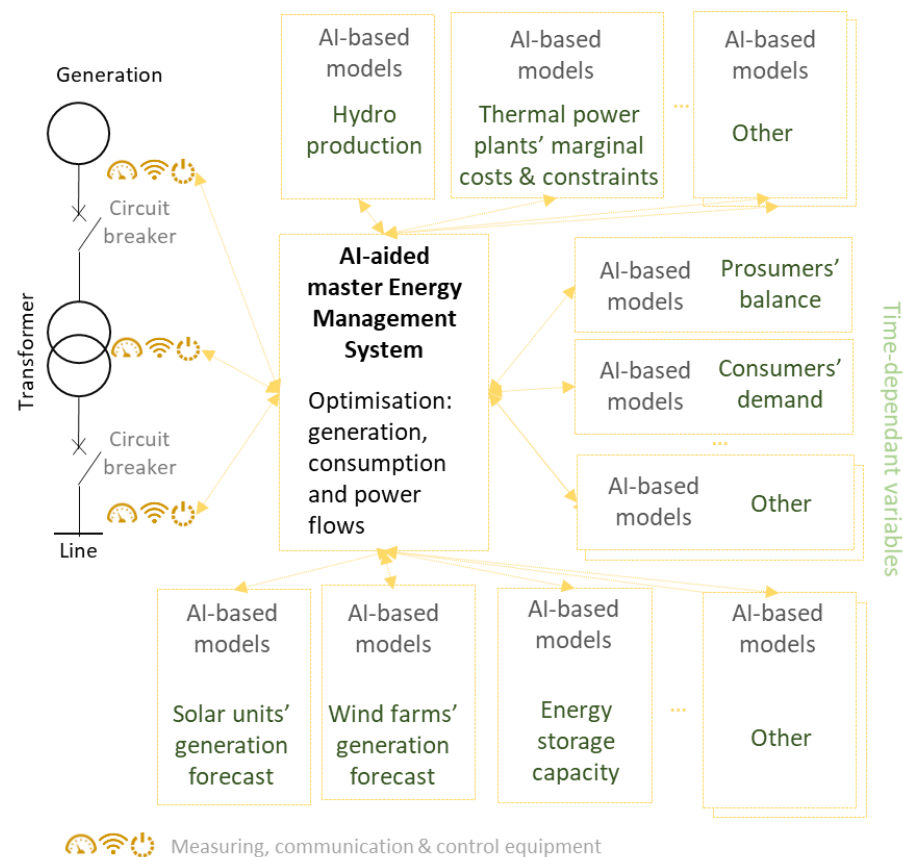


Figure 6. AI-aided Energy Management System.

AI is also heavily researched regarding the topic of predictive maintenance in the power sector [122]. Predictive maintenance of smart grids [123], renewable energy systems [124], photovoltaic systems [125], wind farms [126,127] and distribution networks [128] is explored. In addition to the above-mentioned fields of research, there are numerous other AI-related topics regarding the power sector. AI-based solutions are explored regarding RES sizing options [129], fault detection [130,131], integration of RES [132] and energy efficiency [133].

3.3. Services

Observing literature regarding AI-aided services to consumers in the power sector, an evident gap can be noticed. Other sectors, particularly tech, media and telecom (TMT) and software have done two major things: (1) shifted their focus to being customer-centric companies that aim to provide a premium service [134] and (2) adopted an open innovation business model in which they create to facilitate the co-creation of value for their consumers [135]. Customer engagement via interactive AI platforms is a fast-growing field [136]. A recent study predicts that by 2025, 95% of consumer interactions might be

powered by AI [137]. Although this could be hard to achieve, it is certainly an indication of where AI-aided customer engagement is heading. Businesses will have to adapt to the new, service-orientated reality or will risk becoming uncompetitive in the market. Incorporating AI-based customer support platforms is the first step towards creating an open ecosystem able to foster value co-creation. The key will be to engage customers by actively managing their expectations.

Electric utilities are having a difficult time switching their business models to being customer-centric [138] and are finding it even more difficult to create successful ecosystems. In other words, they keep offering a commodity instead of a service. Following suit, research on AI-aided services in the power sector is limited compared to other sectors [139]. Current research on the topic focuses on analysing active consumer participation in smart energy systems [140], assessing service quality and customer satisfaction [141] and analysing customer experience with chat boxes [142]. AI-aided customer relationship management (CRM) will help improve the ability to predict customer value and will improve the treatment of customers.

4. AI Applications in the Power Sector

Everything in AI is about prediction. In one way or another, AI is used to predict the future. It does so by comparing a string of predetermined conditions with two sets of data: past state and current conditions. The changes in data that have occurred during the time interval observed are used to determine the future state of the system. Given a specified set of parameters, AI generates actions to be taken to ensure optimal operation during this future interval. In the power sector, these actions are applied for various purposes, for power flow optimisation, predictive maintenance, security enhancement, customer support etc.

Applying AI-based solutions requires the use of a considerable amount of data. The term big data simply refers to extremely large datasets. Quality data has always been difficult to attain, but with the digitalisation of the energy system, an increasing amount of data is now being generated. According to a report by Cisco [143], in 2018, an average of 5 exabytes (10^{18} bytes) of data was generated each day. The power and scale of digitalisation are bringing our world closer together, and with increased interconnections and an exponentially larger number of smart devices, the generation of data will only become more significant. Some estimates predict that by 2025 over 460 exabytes of data will be generated each day [144]. When this abundance of information is paired with the increased processing power of today's computers, an ideal foundation for the application of AI is created. As far as the energy sector goes, there will be several major new sources of data: smart meters installed at consumer, storage and producer locations, electric vehicles and numerous smart devices. The Internet of things (IoT) revolution is knocking on the doors of the energy industry, with estimates predicting growth from 25 billion IoT devices that exist today to 75 billion in just a five-year time frame [145].

Figure 7 presents a basic principle of the transactive energy management model where each physical asset has a corresponding virtual counterpart [135]. Digital twins such as the one presented transact in digital space while recording power flows that occur in the system. The key enabler of such a system is the existence of a digital platform. As shown in Figure 7, this platform would consist of three main layers: physical, infrastructure and business. These layers are further divided into six additional interoperability layers: business, functional, information, communication, integration and asset. The aforementioned layers are built upon the physical infrastructure of the power grid and enable a fully-digital peer-to-peer transaction network.

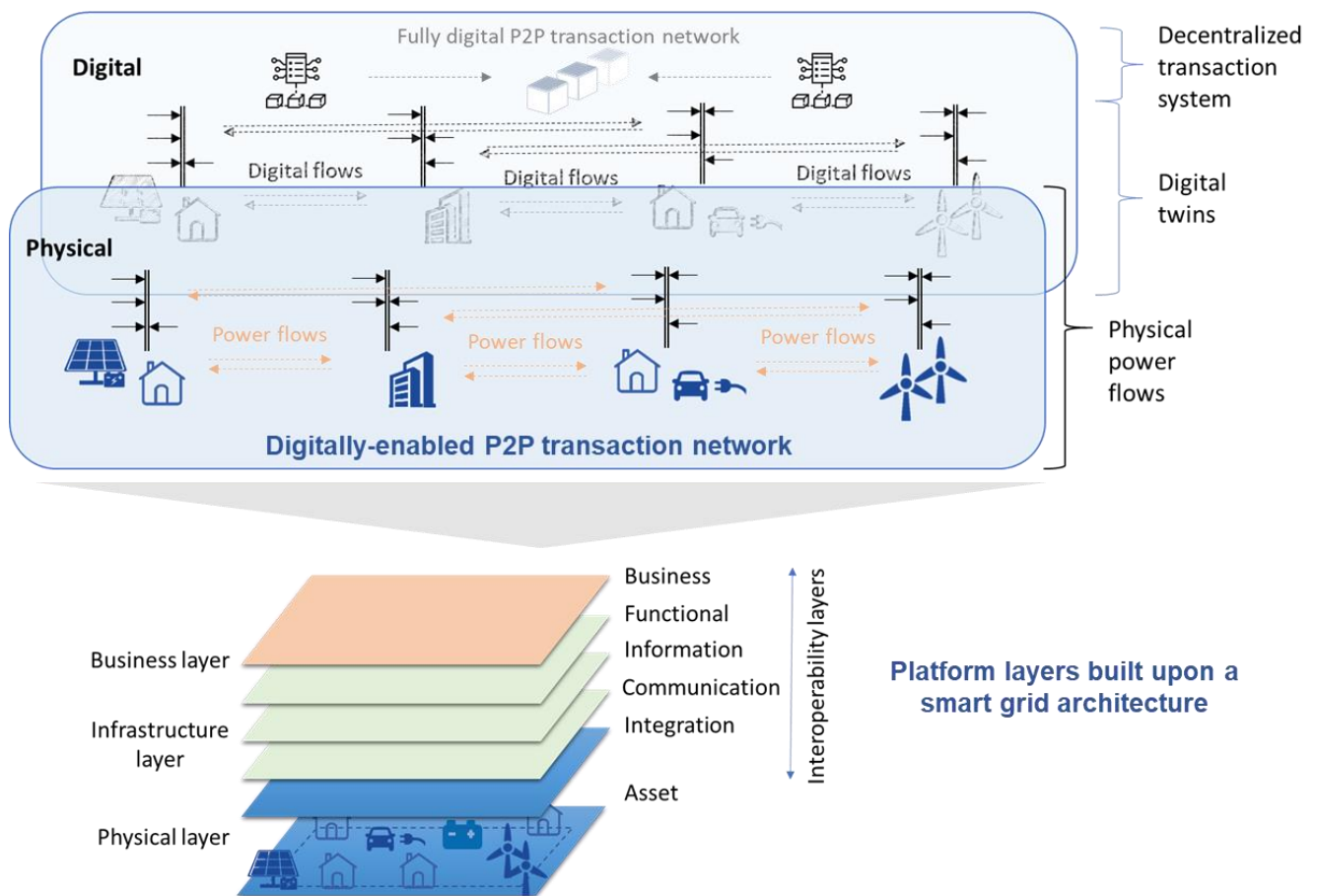


Figure 7. Digitally-enabled transaction network as a basic layer of the smart grid platform.

Power Sector's AI Adoption Rate

As can be seen from the examples of early adopters, AI is destined to be one of the key enablers of the next wave of digital disruption. Facilitated by the rise of digital platforms and by tech giants such as Google, Facebook and Baidu, investments in AI are growing at a rapid pace. According to a recent report by McKinsey, in 2016, companies around the globe invested between USD 26 and 39 billion in AI [146]. The majority of the funding came from tech giants who invested between USD 20 and 30 billion in research, development and deployment of AI systems. Outside the tech world, the adoption of AI is still in its early stages. Energy companies basically shun away from it despite the myriad of advantages AI might bring to the table. The lethargic approach to developing and adopting new solutions is one of the reasons why most utilities have been underperforming in financial markets [147]. Further neglecting the digital reality will only increase the gap energy utilities face when compared to other businesses, such as software or the tech, media and telecom (TMT) industry [135]. However, despite certain headwinds and the slower rate of adoption, AI-aided solutions in the power sector are a rapidly growing field with different interest areas. Figure 8 reveal major areas of AI application in the power sector.

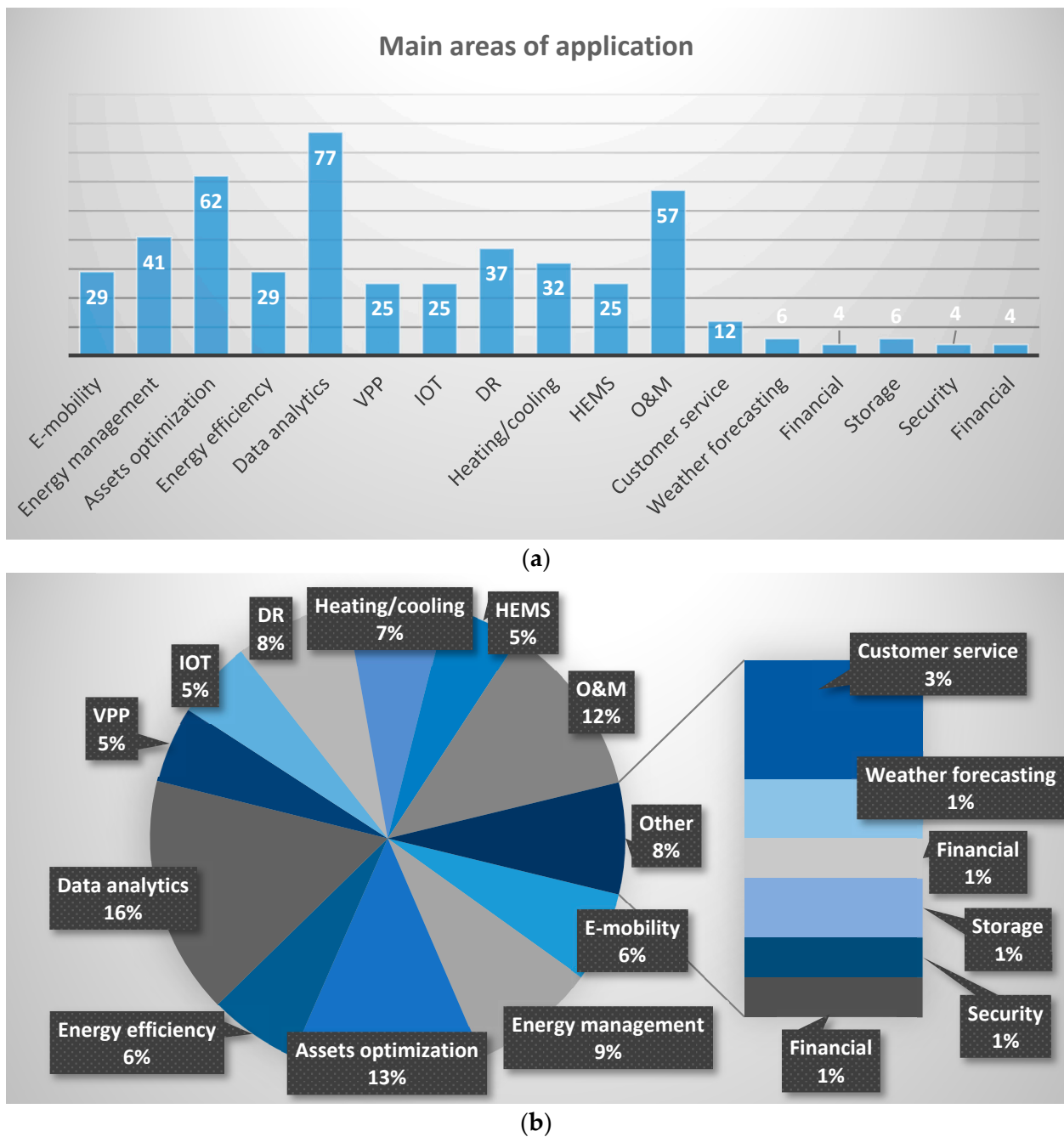


Figure 8. (a) Main areas of application of AI companies in the power sector. (b) Main focus areas of AI companies in the power sector.

Currently, the biggest focus is directed towards four major fields of interest:

- (1) **Optimisation of assets.** According to our research, the majority of AI-based companies in the power sector focus on the application of various optimisation techniques. This field refers to the operational optimisation of particular assets within the system, such as renewable power plants, battery systems, buildings' energy systems and/or home energy management systems. Companies developing applications in this area rely on business models that offer value propositions to residential and commercial users and power plant operators. It should be noted that this field of application does not only deal with power assets. A number of companies analysed use their solutions to integrate home or to build HVAC systems into their optimisation processes. In such a way, they offer an integrated solution to homeowners and companies.

- (2) Optimisation at the system level. This field refers to the management of power flows of the grid and tuning the supply-demand balance by adjusting the production of the generation portfolio, optimally utilising energy storage capacities and applying available demand response schemes. Companies that focus on system-level optimisation generally aim at offering their services to grid operators, utilities and power producers. Grid operators use AI-aided solutions to forecast renewable generation and energy demand and then use this data to optimise their dispatching schedules, and grid power flows. Power generation companies form virtual power plants through which they optimise their production schedules and minimise their exposure to market risk.
- (3) Data analytics. Data analysis and forecasts present the third largest field of AI application in the power sector. Naturally, all AI-based solutions require the use of some form of data analysis. However, companies listed as “optimisers” in the above two fields analyse data to form actions that automatically influence power assets. Companies in the segment of “Data analytics” form insights that are then used for further analysis and form a base for the decision-making process but do not autonomously optimise infrastructural assets. Companies focused on data analytics generally offer their services to utilities and the commercial sector via various monitoring platforms and to power generation companies and system operators that use data analytics in forecasting renewable generation and energy demand.
- (4) Operation and maintenance. O&M is a very common form of AI application in the power sector. Companies that deal with predictive maintenance generally offer their services to power generation companies and the industrial sector. The majority of use cases regarding this field refer to AI-aided analysis of drone or satellite imagery. This field of application also refers to the robotisation of the maintenance process, where (for instance) robots are used to clean PV panels or inspect wind turbines.

Apart from the applications listed in the four fields above, there are several other that have growth potential. The most prolific of these applications are the ones focused on the field of electric mobility. There are numerous companies that use AI-aided solutions regarding electric vehicles. However, in this research, we have considered the ones that focus on applications with a direct influence on the power grid. These generally include companies that develop platforms for the optimisation and/or integration of charging infrastructure and that analyse how to use renewable energy for the charging process. Companies that focus on autonomous driving, upgrading EV efficiency and/or improving battery life have been disregarded. Despite not making it into the top four applications, E-mobility is already on its way to widespread adoption. Looking at AI companies tied to the power sector, we believe that the future of the automotive industry is electric. However, this is a topic for another paper. Sixth on the list seems to be the application of customer service platforms. These help energy providers to understand and serve their customers, offering them a premium customer journey. In this context, the majority of companies use AI to unlock meter-level data, provide consumer insights and make business processes automated and smarter for energy retailers, utilities and grid operators.

Apart from the aforementioned fields, and as witnessed by our research, other popular areas of AI application in the power sector are cybersecurity and financial services. Cybersecurity is a key prerequisite for the enablement of AI. The increased data flows will inevitably raise associated security risks. As the future smart grid will become increasingly distributed, and as it will consist of countless IOT components and smart appliances, so will the entry points for malicious attacks increase. Applying AI-based architecture such as artificial neural networks (ANNs) is proving to mitigate the cybersecurity susceptibilities of the power system. Finally, the financial service segment of AI-aided applications in the power sector will ensure that distributed sources are more easily financeable. At present, around 30–40% of rooftop solar costs refer to soft costs. Aiding the design process and offering a financing solution can prove crucial in the more widespread adoption of solar rooftop solar panels, whether regarding households or commercial users. Figure 9 shows the spatial distribution of analysed companies.

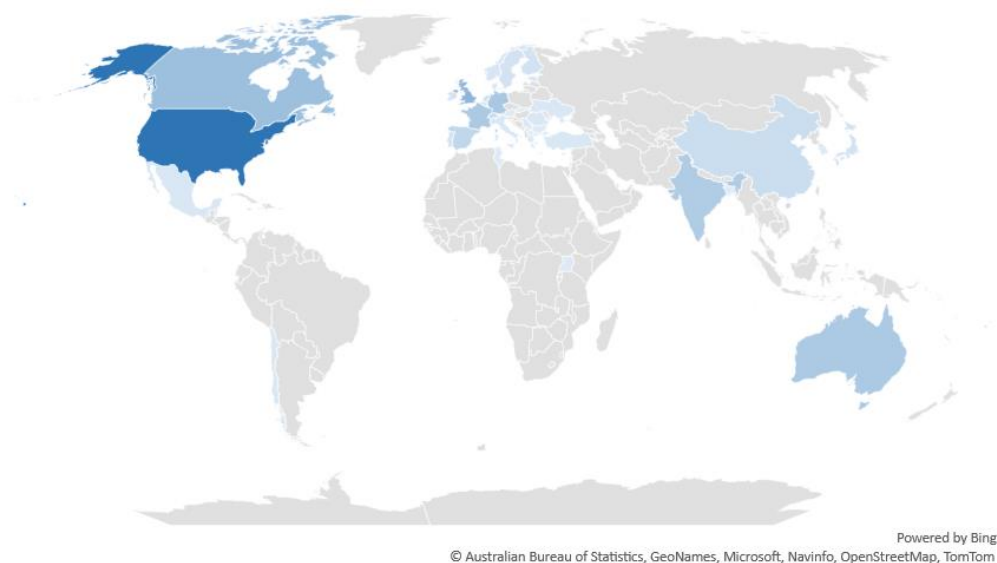


Figure 9. AI-based companies in the power sector across the Globe.

5. Conclusions

The surge of distributed energy sources is causing fundamental changes to the traditional, centralised flow of energy in the power system. Variable renewable energy sources such as wind and solar are disrupting both the energy supply chain and the business models of the companies in the power sector. The increasing amount of energy being produced by these variable sources, paired with the gradual inclusion of electric vehicle charging stations and battery systems, significantly complicates the delicate supply-demand balance of power supply. These are the main reasons why we are witnessing a new era in the management of power. The colossal size and the grave importance of the power systems necessitate the need for the automation of energy management processes. This is where AI-aided technological solutions step in.

As evidenced by the work presented in this paper, the surge of variable renewables and distributed energy sources is accompanied by the surge of companies providing various AI solutions that aim to help solve the complex issues of the new power system. There are currently several major fields of AI application in the power sector. In this research, we have divided them into asset optimisation, system-level optimisation, data analytics and forecasts, predictive maintenance, e-mobility, customer support, cybersecurity and financial services. During our research, several hundred companies have been analysed. This resulted in a list of 220 AI-based companies operating in the power sector. To this date, and to the best of our knowledge, the research presented in the paper is the most comprehensive review of the actual, business-based application of AI in the power sector.

Research presented in the paper shows that the growing researcher interest in applying AI-based techniques in the power sector is accompanied by a growing number of companies following suit in real business applications. These companies are trying to implement numerous innovative solutions to all segments of the energy supply chain. In recent years, start-ups have been developing into successful companies. Many successful start-ups and companies have been acquired by incumbent companies in the sector. In such a way, incumbents combine their strengths with the agility of the start-up they acquired. Despite the lack of agility oftentimes seen in companies of the power sector, we are witnessing a new age in which the sector is becoming increasingly open towards new entrants and disruptors. This process is facilitated by structural changes in the power supply chain.

Business reports regarding other business sectors reveal that AI-based applications are most commonly used to support different types of service optimisations. In other words, they are focused on customers. Traditionally, power companies have always been focused on the process of providing a reliable source of energy. While other sectors, such as

technology, media, and telecom (TMT), have shown more agility and switched to being customer-centric, most power companies have continued to maintain their traditional business model. This is the reason why many of them are failing to provide satisfactory financial returns—because they are still focused on providing a commodity, not a service. Research presented in this paper confirms that in the advent of the power sector's digital age, the majority of digitally-equipped newentrants are still focused on augmenting the efficiency of the process rather than the service.

To provide a final conclusion, given the new energy paradigm driven by distributed energy sources, bidirectional flows and variable RES, it is essential to equip the distribution network with sensors that enable the application of data-driven AI techniques. AI will certainly play a major part in the future of the power sector. However, the successful implementation of these new solutions will take a considerable amount of time and significant effort. To follow the trends set by other industries, the power sector will have to shift its current focus from providing commodities to achieving an open innovation, customer-centric business model. Research presented in the paper shows that major steps have been taken, but further progress in this direction is yet to be made.

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Appendix A

Table A1 lists 220 AI companies in the power sector. It categorizes them by name, country of origin, target market and field of interest according to Figure A1.

no.	name	country	website	market:	
				R – residential, C – commercial, E – power company, power producer, utility	
1	Absolar	UK	https://www.absolar.co.uk/		R/C
	Aids solar power plant design, scans sites to establish solar potential				
			DA		
description					
fields of application:					
M E-mobility	EM Energy management system	AO Asset optimization	EE Energy efficiency	DA Data analytics	VPP Virtual power plant
IOT Internet of things	DR Demand response	HVAC Heating, ventilation, and air conditioning	HEMS Home energy management system	O&M Operation & maintenance	Other

Figure A1. Explanation of Table A1.

Table A1. 220 AI companies in the power sector in 2022. (websites last accessed on 5 January 2023).

1	Absolar Aids solar power plant design, scans sites to establish solar potential	UK DA	https://www.absolar.co.uk/				R/C
2	Accenta Optimizes low carbon heating and cooling needs of buildings	France EE	https://www.accenta.ai/accueil			HVAC	C
3	Actility Carrier grade IoT connectivity platform	France AO	https://www.actility.com/	IOT	DR	HEMS	R
4	Adaptricity Cloud-based network analytics platform enables distribution system operators to better understand, operate, and plan their power grid infrastructure using data-driven network analytics	Switzerland EM DA	https://www.adaptricity.com/en/				E
5	AIDI.solar Developer and integrator of solutions for PV asset management and O&M	Ukraine AO DA	https://www.aidi.solar/				E O&M
6	Ambyint Ambyint developed an automation and analytics solution for artificial lift optimization that builds on traditional physics-based techniques	Canada EE DA	http://ambyint.com/				C/E
7	Amelia Creates personalized digital experiences for customers	USA	https://amelia.ai/				E
8	Amog Advanced hydrodynamic analysis, dynamic cable analysis and static cable stability as well as a range of solutions regarding RES	Australia AO	https://amog.consulting/				E O&M
9	Amperon Provides AI-based smart meter analytics for utilities	USA DA	https://amperon.co/				E
10	Annea Provides predictive maintenance solutions for renewable energy assets such as wind turbines, solar farms, and hydropower plants	Germany	https://annea.ai/				E O&M
11	Argentum Real-time monitoring and control of buildings	Canada EE	https://www.argentum.ai/			HVAC	C
12	Ari analytics Solar power plant production forecast and optimization, data-driven EE optimization of buildings	China AO EE DA	https://www.ari-analytics.com/				E
13	Arloid Automation Helps cut energy costs with a solution that automatically adjusts HVAC settings in buildings based on changing environmental conditions	UK	https://arloid.com/			HVAC	C
14	Arundo Analytics The company uses data and predictive solutions to reduce maintenance costs, improve revenue, and avoid outages for industrial companies	USA	https://www.arundo.com/				C O&M
15	Aurora Solar Develops cloud-based software that aids solar PV engineering design	Canada DA	https://aurorasolar.com/				R/C
16	AutoGrid AutoGrid integrates all distributed energy resources by the use of flexibility management	USA EM VPP	https://www.auto-grid.com/		DR	HEMS	R
17	BeeBryte EE for industrial cooling and HVAC	Singapore AO EE	https://www.beebryte.com/		DR	HVAC	C
18	Beijing Rongxing Technology Develops predictive maintenance solutions, and integrated EMS platforms	China EM VPP	https://www.mixislink.com/				C/E O&M

Table A1. *Cont.*

19	Beyond Limits Beyond Limits is an Industrial and Intelligence capabilities EM	USA	http://www.beyond.ai/ Enterprise grade AI technology company that covers the full range of Artificial Intelligence capabilities EE VPP	C/E
20	Bidgely Customer service platform provider, EM	USA AO	https://www.bidgely.com/ energy analytics for distribution system management related programs EE DA DR	R/C HEMS
21	Blink Energy Inc. Develops a platform that provides utilities with real-time data collection, fault indication, and helps prevents ice on power lines	USA	https://blinkenergy.co/	E O&M
22	BlueWave Blue Wave AI Labs is using predictive analytics to operate nuclear reactors across the United States and to help them operate as safely and efficiently as possible	USA	https://www.bluewaveailabs.com/ DA	E O&M
23	BluWave-ai BluWave-ai uses artificial intelligence to improve operations for energy grids and renewable energy sources such as wind and solar farms	Canada AO	https://www.bluwave-ai.com/	E O&M
24	BluWave-ai Energy optimization for smart grids and fleet electrification M	Canada AO	https://www.bluwave-ai.com/	C/E
25	Bolt Offers electric vehicle charging management solutions M	India	https://bolt.earth/	C
26	BrainBox AI Developed a fully autonomous commercial HVAC solution, BrainBox AI, that uses predictive and self-adapting AI to optimise a building's HVAC system for maximum energy reduction	Canada	https://brainboxai.com/en HVAC	C
27	Bueno Analytics and optimization solutions for building management AO	Australia AO	https://www.buenosystems.com.au/ DA HVAC	C
28	Buzz Solutions Buzz Solutions is developing artificial intelligence technology to spot and analyze power line flaws so companies can repair them before a fire starts	USA	http://buzzsolutions.co/	E O&M
29	C3 AI Uses machine learning techniques to enable accurate forecasting, benchmarking, building optimization, demand response, and anomaly detection. The company's digital platform for utilities offers application for performance optimization for gas power plants, efficiency optimization of heating and cooling plants, energy analysis and management for enterprise and consumer customers, and anomaly detection of wind turbines.	USA AO	https://c3.ai/ EE DA VPP IOT DR HVAC	C/E O&M
30	Carbon Relay Carbon Relay tackles data centre cooling with AI	USA	https://www.carbonrelay.com/ EE HVAC	C
31	Centrica Centrica's REstore provides cloud-based demand side management. The technology is currently used by more than 150 industrial energy consumers in Europe.	Belgium	https://www.centricabusinesssolutions.com/energy-solutions/products/energy-optimisation-solutions?redirect=restoreeu DR	C

Table A1. Cont.

32	ChargePoint Kisensum develops software to control and optimize energy resources, including photovoltaics, energy storage and charging stations for electric vehicles	USA	https://www.chargepoint.com/en-gb				C
33	CIM Data-driven building operations software	AO Australia	https://www.cim.io/			HVAC	C
34	Circunomics Optimizing second life and recycling of batteries	Germany AO	https://www.circunomics.com/				C/E
35	Cleandrone Develops autonomous drones for thermal imaging and cleaning of solar panels and glass surfaces	Spain	http://www.cleandrone.com/				E O&M
36	CLEAResult The largest provider of emission-reducing energy solutions across North America	USA EM	https://www.clearesult.com/	EE	DR		R HEMS
37	Cleveron Building EMS	Switzerland	https://cleveron.ch/		IOT	HVAC	C
38	ClimaCell Delivers weather insights helping to manage weather related challenges	USA	https://www.climacell.co/				C/E
39	Climatik Meteorological data analysis	Mexico	https://climatik.net/	DA			C/E
40	Clobotics Provides drone-based wind turbine inspection and monitoring repair and maintenance	China	https://clobotics.com/				E O&M
41	COI Energy Services COI Energy Solutions deals with energy waste issues through its EMS platform improving energy performance of buildings	USA	https://www.coienergyservices.com/	EE		HVAC	C/E O&M
42	Coulomb AI Optimizes batteries' operation	USA AO	https://coulomb.ai/				C/E
43	cove.tool Provides data, automation, and collaborative tools to design better buildings	USA AO	https://www.cove.tools/				C
44	Crusoe The company's behind-the-meter load approach enables alternative revenue streams for renewable and clean energy projects	USA AO	https://www.crusoeenergy.com/				E
45	cyberGRID cyberGRID offers innovative ICT-based flexibility management technology, integration of renewable energies and storage devices	Austria EM	https://www.cyber-grid.com/	AO	VPP	DR	C/E
46	Dabbell DABBEL controls the energy system in buildings reducing energy consumption	Germany EE	https://www.dabbell.eu			HVAC	C O&M
47	dcbel Develops AI-driven sustainable technologies which enable people to leverage solar energy to power their cars	Canada M	https://www.dcbel.energy/				C

Table A1. *Cont.*

48	DCbrain	France	https://dcbrain.com/					C/E
	Detects anomalies and anticipates incidents, and uses data to predict future requirements							
	DA							
49	DCSix Technologies	Ireland	https://www.dcsixtechnologies.com/					R/C
	Provides an energy monitoring platform that help reduce energy consumption							
	HEMS							
50	Deepmind	UK	https://deepmind.com/					C/E
	Google's DeepMind is the world leader in artificial intelligence research and its application in different fields, such as games, medicine, energy efficiency.							
		AO	EE	VPP	IOT	HVAC		
51	DeepVolt	Tunisia	https://deepvolt.io/					C
	AI-powered software for the placement and sizing of electric vehicle charging stations							
	M							
52	Detect Technologies	India	https://detecttechnologies.com/					C/E
	Identifies fault differences, workforce safety hazards, monitors process automation and possible risks, generates analytics on data consumption							
			EE	IOT				
53	Dexter Energy	Netherlands	https://dexterenergy.nl					C/E
	Dexter provides forecasting and dispatching solutions based on AI and cloud-based technology that increases efficiency and reduces cost							
		EM	DA					
54	Dynamhex	USA	https://dynmhx.io/					E
	Customer service platform provider, energy analytics							
	DA							
55	EasyMile	France	https://easymile.com/					C
	Develops autonomous driving systems and smart mobility solutions							
	M							
56	Ecolibrium Energy	India	https://www.ecolibrium.io/					C
	The company's platform provides a holistic view of buildings' energy profile, develops customer engagement systems							
			EE	DA	IOT			
57	Ecotropy	France	https://ecotropy.fr/					C
	Provides analyses of buildings' energy performance and optimizes energy retrofit of buildings							
			HVAC					
58	Effenco	Canada	https://www.effenco.com/					C
	Develops electric hybrid systems for electrification of transport and electric powertrains							
	M							
59	ei	Portugal	https://www.galpsolar.com/pt/					E
	Develops solar energy monitoring solutions							
	O&M							
60	Elevate	North Macedonia	https://elevate-global.biz/index.html					E
	Develops autonomous energy forecasting solutions for utilities							
	DA							
61	eleXsys	Australia	https://ellexsys.com/					E
	Develops AI-driven smart grid solutions							
	EM							
62	Emuron	India	https://www.emuron.com/					C
	Develops IoT-enabled battery management systems							
	M							

Table A1. Cont.

63	Enercast Forecasts energy production	Germany	https://www.enercast.de/					E
			DA					
64	Energency Provides a platform for enterprises to optimize their EE	France	https://www.energency.com/					C
			EE DA					
65	Energy Pool Builds and operates demand-side management solutions	France	https://www.energy-pool.eu/en/					C/E
	EM		VPP			DR		
66	Energy X Developed an AI-driven platform that allows corporate and individual users to invest in renewable energy projects worldwide	South Korea	http://www.energix.co.kr/					C
67	Enerlogix Developed a platform offering HVAC management solutions for commercial applications	Spain	https://enerlogix.es/					C
						HVAC		
68	Enervalis Developed a platform optimizes homes, buildings and electric vehicle charging stations	Belgium	https://enervalis.com/					R/C/E
	M EM AO		DA VPP IOT DR				HEMS	
69	Energyield Develops solutions to help with fault forecasting, identification and localization, life-time estimation, energy loss estimation and grid health assessment	Sweden	https://www.energyield.com/					C/E
							O&M	
70	Enfor Develops software solutions for power generation forecasting, optimizes district heating systems	Denmark	https://enfor.dk/					E
	M		DA			DR		
71	EnPowered Facilitates companies to invest in energy efficiency solutions	Canada	https://enpowered.com/					R/C
	EM		EE DA VPP			DR		
72	Entelios EMS and trading	Norway	https://www.entelios.com/					R/C
	EM					DR		
73	EQuota Energy AI & Big Data supported EMS Service provider	China	https://equotaenergy.com/en/					C
		AO	DA					
74	Eve Analytics solutions and emissions reporting to empower corporate fleet electrification	Ireland	https://evE-mob.io/					C
	M	E-M						
75	Evio Develops e-mobility solutions	Portugal	https://go-evio.com/					C
	M							
76	Evolve Energy Developed the Kraken platform for customer management now licensed to E.ON, Npower, Origin Energy and EDF Energy and providing support to millions of customers. The company is also focused on providing flexibility and market access.	USA	https://www.evolveenergy.com/					E
	M					IOT DR		
77	Exergenics Analytics solutions for building management	Australia	https://www.exergenics.com/					C
		AO	DA					
78	Flexitricity Largest demand response aggregator in UK	UK	https://www.flexitricity.com/					C/E
	EM					DR		

Table A1. Cont.

79	Flutura	India	http://www.flutura.com/	E
	Developed a platform Cerebra focused on improving asset uptime and operational efficiency	AO		O&M
80	Foghorn Systems	USA	https://www.foghorn.io/	C/E
	FogHorn is an IoT platform that provides a complete edge solution consisting of a highly miniaturized complex event processing engine that derives real-time insights	DA	IOT	
81	Fresh Energy	Germany	http://getfresh.energy/	R/C
	Develops online building EMS solutions		IOT	HVAC HEMS
82	Fulcrum3D	Australia	https://www.fulcrum3d.com/	E
	Remote sensing, forecasting, data capture, and reporting for solar and wind power	DA		
83	Future Grid	Australia	https://future-grid.com/	E
	Real-time smart meter data analytics	DA		
84	Gaiascope	USA	https://gaia-scope.com/	E
	Provides forecasting software for energy storage systems and empowers customers to achieve bid and trading optimization for their power assets	AO	DA	
85	Gbatteries	Canada	https://www.gbatteries.com/	C/E
	Optimizes battery charging solutions for electric vehicles	M		
86	Generac Grid Services	Canada	https://www.generac.com/	E
	Manages grid assets and balance supply and demand in real time	EM	VPP	DR
87	GetJenny	Finland	https://www.getjenny.com/	E
	Develops self-service solutions			
88	Gilytics AG	Switzerland	https://www.gilytics.com/	E
	Automates infrastructure planning, routing, and monitoring	DA		
89	Glint Solar	Norway	https://www.glintsolar.ai/	E
	Identifies and analyses optimal solar sites using satellite imagery	DA		
90	Green Running	UK	https://verv.energy/	R
	Predictive maintenance technology for smart, sustainable appliances through Verv, a home energy assistant which monitors energy use in a home to help cut costs and save energy			
91	Greenbird	Norway	https://www.greenbird.com/	E
	Big Data integration for utilities	DA		
92	GreenPocket	Germany	https://www.greenpocket.com/	E
	Develops an energy management system and visualization software			
93	GreenWhale	Germany	https://thegreenwhale.com/	C
	Operates private and public vehicle-to-grid charging infrastructure	M		
94	Grid AI	Japan	https://gridpredict.jp/	E
	Grid optimization	EM		

Table A1. Cont.

95	GridBeyond Uses AI-powered technology to optimize system operation to facilitate savings and manage price volatility	UK M EM	https://gridbeyond.com/ VPP DR	C/E
96	GridIMP AI driven fully automated technology takes care of day-to-day EMS decisions	UK EM AO	https://gridimp.com/ VPP IOT DR	C HEMS
97	Gridium The company's data platform automatically aggregates energy interval data, billing data, weather history data, and weather forecast data to power a rich set of analytic services	USA DA	http://www.gridium.com	C
98	Hank Hank is enabling service providers and building owners to proactively manage HVAC costs and improve EE of commercial buildings	USA EE	https://www.hank.re/ HVAC	C
99	Hive Power The company's platform monitors resource distribution and analyzes the usage behavior of consumers (from energy communities to EV charging), it also provides tools to optimize energy trading and prevent grid overloads	Switzerland M EM AO	https://www.hivepower.tech/ DA VPP	C/E
100	Homeys Delivers a platform that connects sensors and provides real-time data and performs analysis of heating settings and energy consumption	France HVAC	https://www.homeys.io/	C
101	Homi Provides home automation solutions for residential buildings	China HEMS	https://myhomi.io/	R/C
102	inbenta Automate customer service	Spain E	https://www.inbenta.com/en/	E
103	IND Technology IND.T is uses DA and smart sensing to reduce unplanned outages	Australia DA	https://ind-technology.com/	E O&M
104	Informetis Develops energy related services using Big Data and machine learning in order to improve EE	Japan EE	https://www.informetis.com/	C
105	Infra Solar Focuses on digitizing electricity consumption, optimising electric mobility, and energy consumption management	Netherlands DA	https://infrasolar.com.br/	E
106	Innowatts Innowatts' platform transforms how energy providers understand and serve their customers. It uses AI to unlock meter-level data, provide consumer insights and make business processes automated and smarter for energy retailers, utilities and grid operators.	USA DA	http://www.innowatts.com/	E
107	Instylesolar Solar power analytics	Australia DA	https://instylesolar.com/	R/C
108	IntelliView Leak detection and video analytics systems for surveillance	Canada O&M	https://www.intelliviewtech.com/	C/E
109	Invenia Invenia's Energy Intelligence System is a cloud-based machine learning platform that uses big, high frequency data to solve complex problems in real time. Invenia currently applies its platform to optimize electric utility operations as well as electricity markets themselves.	Canada EM	https://www.invenia.ca/	E

Table A1. Cont.

110	Invenia Forecasting and pattern recognition to manage and optimize operations EM	Canada AO	https://invenia.ca/ DA	E
111	Ion Energy Labs Develops battery management and mobility systems M	India AO	https://www.ionenergy.co/	C/E
112	IONATE Real-time monitoring and control of the electricity grid-edge EM	UK	https://www.ionate.energy/	E
113	Itron Itron's portfolio of smart networks, software, services, meters and sensors helps our customers better manage energy and water EM	USA AO	https://www.itron.com/ DR	C/E
114	Jungle AI AI-based Canopy aims to increase production and prevent unplanned downtime and has industrial and power generation applications	Portugal	https://www.jungle.ai/	C/E O&M
115	Kagera AI Optimizes production and minimizes downtime AO	Serbia AO	https://www.kagera.ai/ EE	E O&M
116	Kapacity.io The company provides energy optimisation and demand response services for buildings by connecting to buildings through building management systems and adjusting HVAC energy consumption in real time EE	Finland	https://kapacity.io/ DR	C HVAC
117	Kayrros Kayrros is an advanced DA company that helps global energy market participants make better investment decisions DA	France	http://www.kayrros.com/	C/E
118	Kiwi Power (Engie) Provides asset optimisation using flexibility of generation capacity to earn revenue by accessing wholesale market opportunities and participating in grid demand-side services EM	UK AO	https://www.kiwpowered.com/ VPP	C/E
119	kWIQly AI based search and analytics DA	Switzerland	https://kwiqly.com/	C
120	Leanheat Leanheat aims to use artificial intelligence to improve climate control in multi-family buildings EE	Finland	https://leanheat.com/ IOT	R HVAC HEMS
121	LEBO ROBOTICS Providing smart maintenance for wind power plants	Japan	https://www.leborobotics.com/en	E O&M
122	Levelise Home EMS system that links high numbers of domestic battery systems to a suite of advanced algorithmic controls. By coordinating the actions of these batteries Levelise can provide balancing services to the electricity transmission operator and to suppliers	UK	https://www.levelise.com/ IOT	R DR HEMS
123	LifeSmart The company's platform offers home automation services	China	https://cn.ilifsmart.com/	R/C HEMS

Table A1. Cont.

124	Limejump Platform that delivers flexible energy in real time by an aggregation of flexible energy generation and storage assets. EM	UK	https://www.limejump.com/ DR	C/E
125	Lition Lition and Watty provide a power usage monitoring system for commercial or personal purposes. AO	Sweden	https://lition.io/ IOT	R HEMS
126	LiveEO Provider of satellite-based power grid monitoring services	Germany	https://live-eo.com/	E O&M
127	Loggma Data monitoring and analysis for solar power plants DA	Turkey	https://loggma.com.tr/en/	E
128	LogicLadder Develops an EMS and monitoring platform EM	India	https://www.logicladder.com/ DA	C
129	MAC MAC created GridWatch - low-cost earth fault monitoring product solutions based on cloud back-end solution IOT	Ireland	http://www.mac.ie/Utilities/EarthFault.aspx	E O&M
130	Mandulis Energy Develops and operates renewable energy projects focused on the sustainable biomass sector AO	Uganda	https://www.mandulisenergy.com/ IOT	E HEMS O&M
131	Meteo-Logic They are a DA company focusing on weather prediction and how it impacts the energy commodity market. The company uses machine learning and big data to create self-learning algorithms that produce accurate predictions when it comes to weather and its impact on energy supply. The algorithm adapts and adjusts to the data that comes in to predict future behaviour as well.	Israel	http://meteo-logic.com/	E O&M
132	METRON Metron's platform provides insight into energy usage and provides data on energy intelligence strategies DA	France	https://www.metronlab.com/	C
133	Mindtitan Customer service, demand forecast, optimization of energy production and scheduling, defect detection EM	Estonia AO	https://mindtitan.com/	E O&M
134	Mobilyze Provides a location intelligence platform to identify EV charging hotspots M	Slovakia	https://mobilyze.it/#/	C
135	Moduly Moduly helps residential and commercial electricity users optimize their EE by allowing them to shift consumption peak hours, and reduce energy consumption	Canada	https://moduly.io/ IOT	R/C HVAC HEMS
136	moixa Focuses on smart battery systems and smart charging, and platform that allows households to minimize energy bills by optimizing battery and electric vehicle performance M	UK AO	https://moixa.com/	C HEMS
137	Morgan Solar Inc. Solar power plant production forecast and optimization, data-driven EE optimization of buildings AO	Canada	https://morgansolar.com/ DA	R/C HVAC
138	Myst AI Myst AI is a developer of the AI-based data analysis platform intended for electricity demand and supply forecasting DA	USA	https://www.myst.ai/	E

Table A1. Cont.

139	NanoLock Security	USA	https://nanolocksecurity.com/					C/E
	NanoLock device-level protection and management secures IoT and connected devices							
140	NEC	Japan	https://www.nec.com/					C/E
	Focuses on plant failure prediction systems and building EMS systems							
						HVAC	O&M	
141	Nest Labs	USA	https://nest.com/					R
	Nest Labs is a home automation company manufacturing sensor-driven, Wi-Fi-enabled, self-learning thermostats and smoke detectors							
			EE		IOT	HVAC	HEMS	
142	Neurons Lab	Ukraine	https://www.neurons-lab.com					C
	Neurons Lab specializes in designing AI-based solutions to maximize energy outputs for small businesses and start-ups by forecasting renewable energy production (PV, Wind) and predicting energy consumption							
		AO	DA					
143	Nnergix	Spain	https://www.nnergix.com/					C/E
	Manages renewable energy and optimizes spinning reserves by AI-based algorithms and analytic models. Incorporates artificial intelligence to improve the accuracy of renewable energy forecasting							
		AO	DA	VPP				
144	Notilo Plus	France	https://notiloplus.com/					E
	Provides underwater analysis							
								O&M
145	NRGI.ai	Ireland	https://www.nrgi.ai/					C
	Nrgi is developing an AI-based B2B energy marketplace with energy price benchmarks, using an AI-based forecasting engine							
			DA					
146	Octopus Energy	UK	https://octopus.energy/					R/C
	Developed a cloud-based smart grid platform that balances loads around the grid							
	M	EM		VPP		DR		
147	Ogre AI	Romania	https://www.ogre.ai/					C/E
	Ogre AI develops automated B2B platform which uses machine learning to offer financial and operational decision-making support to energy and utilities companies							
			DA					
148	Open Energi	UK	https://openenergi.com/					R/C/E
	The company's platform, Dynamic Demand 2.0, automatically optimizes and trades power from low carbon technologies such as battery storage, EVs, demand-side response, hydrogen electrolyzers and smart buildings							
	M	EM	AO			DR	HEMS	
149	Origami Energy	UK	https://www.origamienergy.com/					E
	Origami's independent energy data platform enables real-time market access, physical control, scheduling & dispatch, contract & position management, and price & volume forecasting							
		EM	DA					
150	Orison	USA	https://orison.com/					R/C
	Orison offers residential and commercial self-installable energy storage systems paired with flexible fleet control tools that enable electric utilities and energy retailers to improve grid resilience and manage peak loads							
		EM			IOT	DR		
151	Osperity	USA	https://osperity.com/					C
	Osperity's technology provides AI-driven intelligent visual monitoring for industrial operations that can result in improved safety, reduced carbon footprints, etc.							
			EE DA					O&M

Table A1. Cont.

152	Palmetto Clean Technology	USA	https://palmetto.com/	R
	Insights and design for rooftop solar installation, monitoring and predictive maintenance			O&M
	DA			
153	Piclo	UK	https://www.piclo.energy/	R/C
	Operates a cloud-based platform and integrated service that supports the end-to-end process of procuring and operating flexibility			
	EM	AO		DR
154	Ping Services	Australia	https://ping.services/	E
	Ping records changes in this signature to continuously monitor the health of wind turbine blades and use advanced acoustic analysis to detect damage			O&M
155	Ping Things	USA	https://www.pingthings.io/	E
	The company uses big data software and science to detect events on the power grid. The platform records data from internal and external sources to utilities. This helps predict if an event, such as asset failure, will occur and alert the control room so they can avoid any outages.			
	AO			O&M
156	Plexflo	USA	https://www.plexflo.com/	C
	Provides operational steps for EV charging station development, and simulates the electric grid to provide energy DA			
	M			
157	PowerPeers	Netherlands	https://www.powerpeers.nl/	R/C
	Aggregates smaller producers and consumers			
	AO			DR
158	Prescinto	India	https://prescinto.ai/	E
	The company's platform offers advanced DA, enhances the performance of renewable energy or energy storage assets, streamlines operations and maintenance			
	EE DA			O&M
159	Proasistech	Spain	http://bd4bs.com/	E
	Optimizes operation of generation assets			
	AO			
160	Quadrical.Ai	Canada	https://www.quadrical.ai/	C/E
	Quadrical develops AI-based solar plant monitoring and forecasting platform			
	DA			O&M
161	Quodus	Spain	https://quodus.ai/	E
	Developed a platform for infrastructure monitoring, optimization of energy consumption, and analytics			
	AO		DA	
162	R8tech	Estonia	https://r8tech.io/	C
	R8 Digital Operator developed add-on software for existing building automation systems			
	EE		HVAC	O&M
163	Ramanujan	India	http://ramanujaninc.com/	R/C/E
	Designs, develops and operates service-oriented solutions that enhance utility performance and consumer EMS			
	EMS		IOT DR	HEMS
164	Raptor Maps	USA	https://raptormaps.com/	E
	Advanced analytics and aerial thermal inspections for solar power plants			
	DA			O&M
165	Rated Power	Spain	https://ratedpower.com/	E
	Their technology finds the smartest way to maximize the value of photovoltaic plants, scanning millions of iterations, finding the best configuration and immediately generating hundreds of pages of engineering documents			
	AO		DA	
166	Raycatch	Israel	https://www.raycatch.com/	C/E
	Raycatch developed DeepSolar, a digital asset management system that automates and optimizes solar PV assets			
	DA			O&M

Table A1. Cont.

167	Regalgrid	Italy	https://www.regalgrid.com/en/	R
	Works as an aggregator connecting homes to one community where clean energy is produced, stored and distributed among a network of people			
			DR	HEMS
168	Relectrify	Australia	https://www.relectrify.com/	R/C
	Increases longevity of batteries			O&M
169	Rhythmos	USA	https://rhythmos.io/	C/E
	Rhythmos assembles critical information from legacy systems to automatically identify, characterize, quantify and forecast EV charging needs and host utility grid constraints			
	M	AO		
170	Sawatch Labs	USA	https://www.sawatchlabs.com/	C
	EV analytics including tailpipe emissions calculations, vehicle specific EV recommendations, site specific supply equipment, energy demand projections, and operational analysis of EV			
	M		DA	
171	Schneider Electric	USA	http://www.auto-grid.com/	C/E
	AutoGrid Systems by Schneider Electric organizes energy data and employs big DA to generate real-time predictions, and offers flexibility solutions			
	EM		DA	VPP
172	Scopito	Denmark	http://scopito.com/	E
	Predictive maintenance for power lines, and solar and wind power plants			
			DA	O&M
173	Senfal (Vattenfall)	Netherlands	https://group.vattenfall.com/	C/E
	Senfal is a Dutch start-up that offers innovative software services to large industrial customers, wind and solar farms, as well as battery owners for unlocking value from flexibility			
	EM	AO		
174	Sensgreen	Turkey	https://www.sensgreen.com/	C
	Optimizes building management			
		AO		HVAC
				O&M
175	Simerse AI	UK	https://www.simerse.com/	E
	Helping electric utilities find defects on critical utility equipment			
				O&M
176	Skyqraft	Sweden	http://skyqraft.com/	E
	Skyqraft conducts aerial inspection of power lines through unmanned airplanes and machine learning			
				O&M
177	SkyX	Canada	https://skyx.com/	E
	Drone-aided predictive maintenance			
			DA	O&M
178	SmartCat	Serbia	https://smartcat.io	C
	SmartCat develops software used to optimize heating and cooling devices, and to reduce electricity consumption			
			EE	HVAC
179	SmartHelio	Switzerland	https://smarthelio.com/	E
	An end-to-end real-time analytics platform for solar plants			
				O&M
180	Smartive	Spain	https://smartive.eu/	E
	Provides a cloud-based monitoring solution for wind farm assets			
				O&M
181	Sobolt	Netherlands	https://www.sobolt.com	C
	Sobolt operates an AI solution HeatPuls that maps heat losses in buildings			
				HVAC

Table A1. Cont.

182	Social Energy Creates solutions flexibility and demand response for homeowners and businesses AO	UK	https://social.energy/	DR	HVAC	HEMS	R/C
183	Solar AI Aids solar power plant desing, scans sites to establish solar potential DA	Singapore	https://getsolar.ai/				C
184	Solar Captus Their platforms aid designing rooftop solar units, and facilitate digital marketing DA	Australia	https://www.solarcaptus.com/				R/C
185	Solar Inspectron AI Drone thermal inspections of PV assets	Greece	https://solarinspectron.ai/				E O&M
186	Solavio Autonomous solar panel cleaning	India	https://www.solavio.com/				E O&M
187	SOLshare SOLshare created a peer-to-peer energy exchange network of rural households and small businesses with rooftop solar home systems AO	Bangladesh	https://solshare.com/	VPP	IOT		R
188	Solstice The company operates a platform that helps integrate community-based solar power VPP	USA	http://solstice.us				R/C
189	Solytic Provides monitoring solutions of large-scale rooftop and outdoor systems and develops comprehensive databases for the provision of device benchmarks and analysis DA	Germany	https://www.solytic.com/				E O&M
190	SparkCognition Developed cyber-physical software for the safety, security, and reliability	USA	https://www.sparkcognition.com/				C/E
191	Stem The company's platform Athena forms virtual power plants to maximize the value of energy resources through a combination of machine learning and predictive analytics M	USA	https://www.stem.com/	DA	VPP		C/E
192	Sterblue Designed a platform that manages infrastructure inspections through image recognition AO	France	https://www.sterblue.com/		VPP		C
193	suenal Optimizes battery storage AO	Germany	https://suenal.energy/				E
194	sun2wheel Develops V2H and V2B charging systems that ensure the availability of energy in EVs during the day and feeds back the excess to households at night M	Germany	https://sun2wheel.com/en/home/				R
195	Sunai The company developed a platform NEURAL that gives operational guidelines to O&M teams DA	Chile	https://sunai.cl/en/				E O&M
196	Suncast Develops models that perform energy forecasts in photovoltaic power plants and applies AI and ML to estimate soiling in photovoltaic modules DA	Chile	https://www.suncast.cl/				E O&M

Table A1. Cont.

197	Sync Energy	USA	https://www.sync.energy/	E
	Sync develops no-code AI-based predictive simulations and analytics, that streamline planning and operations for electrical utilities.			
			DA	
198	The Solar Labs	India	http://thesolarlabs.com	C/E
	Develops software that aids solar PV engineering design			
			DA	
199	Thermelgy	India	https://www.thermelgy.ai/	C/E
	Enables end users to manage their energy assets more efficiently			
		AO	EE IOT HVAC	
200	ThermoVault	Belgium	https://www.thermovault.com/	R
	Provides an all-in retrofit solution that transforms electrical space and water heaters into energy-efficient storage devices			
			EE DR HVAC	
201	Tibber	Sweden	http://tibber.com	R
	Tibber is a digital electricity supplier that uses AI to regulate power for houses based on their predicted levels of consumption			
		AO	IOT HEMS	
202	tiko	Switzerland	https://tiko.energy/	R/C
	tiko's Virtual Power Plant is designed to stabilize the grid and reduce these fluctuations			
		EM	VPP	
203	TokWise	Bulgaria	https://www.tokwise.com/	E
	Develops algorithms for energy forecasting			
			DA	
204	Trendometric	Romania	https://trendometrics.com/	E
	Develops energy forecast models			
			DA	
205	Twaice	Germany	https://www.twaice.com/	C/E
	Provider of digital twin-based battery DA and management platform			
		AO	DA	O&M
206	Unleash live	Australia	https://unleashlive.com/	E
	Video analytics for predictive maintenance			
				O&M
207	Upside Energy	UK	https://www.krakenflex.com/	E
	Software platform for managing, controlling and optimising all distributed energy resources			
		EM	DA VPP	
208	Urbint	USA	https://www.urbint.com	C/E
	Urbint predicts threats to workers and critical infrastructure to stop incidents before they happen			
				O&M
209	Verdigris Technologies	USA	https://verdigris.co/	C
	Verdigris Technologies is a SaaS-based platform that develops artificial intelligence in order to optimize energy consumption			
		AO	DR	
210	VIA	USA	https://www.solvewithvia.com/	E
	Predicts energy demand, grid power flows, possible outages, and renewable energy production by analysing data collected by smart meters, drones, and sensors			
			DA IOT	
211	Virtual Power Solutions	UK	https://www.vps-energy.com/	R/C/E
	The company's platforms focus on smart homes and smart cities. EMS system that combines demand management with building automation technology targeted for the business sector			
		EM AO	VPP DR HEMS	
212	Visor	Portugal	https://www.visor.ai/	E
	Customer service automation with AI bots			

Table A1. Cont.

213	visualAI Technology integrates data from drones, IoT sensors, and offline sources in predictive maintenance and asset management	India AO	https://futr.energy/	C/E O&M
214	Voltalis Demand response for home heating	France	https://www.voltalis.com/	R DR HVAC
215	VoltPost Offers hardware-as-a-service that transforms lampposts into smart electric vehicle charging stations	USA M	https://www.voltpost.com/	C
216	Wärtsilä Greensmith Energy (Wärtsilä) uses software-based intelligence and machine learning to enhance grid systems and networks	Finland EM	https://www.wartsila.com/ VPP DR	C/E
217	wibeee Offers a home EMS platform	Spain	https://wibeee.com/	R HEMS
218	Willow Willow empowers asset owners and operators to make proactive, data-led decisions in real time	Australia AO	https://www.willowinc.com/	C
219	Wirewatt The company provides solar leases to customers of any business that installs or sells solar panels	Mexico	https://www.wirewatt.com/	R/C
220	Worlds Worlds' technology combines deep learning and IoT inside of a 4D environment to give organizations the ability to observe their organization's physical space and then, analyze and learn from their physical surroundings	USA DA	https://worlds.io/	C/E

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